

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 01

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Recap of Previous Lecture



Topic

Packet Switching

Topic

Virtual Circuit Switching



Topics to be Covered



Topic

IPv4 Address



ABOUT ME



Hello, I'm **Abhishek**

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- 12 years of GATE CS teaching experience

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Topic : IP Address



- Unique identifier
- Assigned to a device that uses Internet Protocol for communication
- Logical Address (IP Address)
- MAC Address (Physical Address) (Hardware Address)
[48-bit size, NIC Card Address]



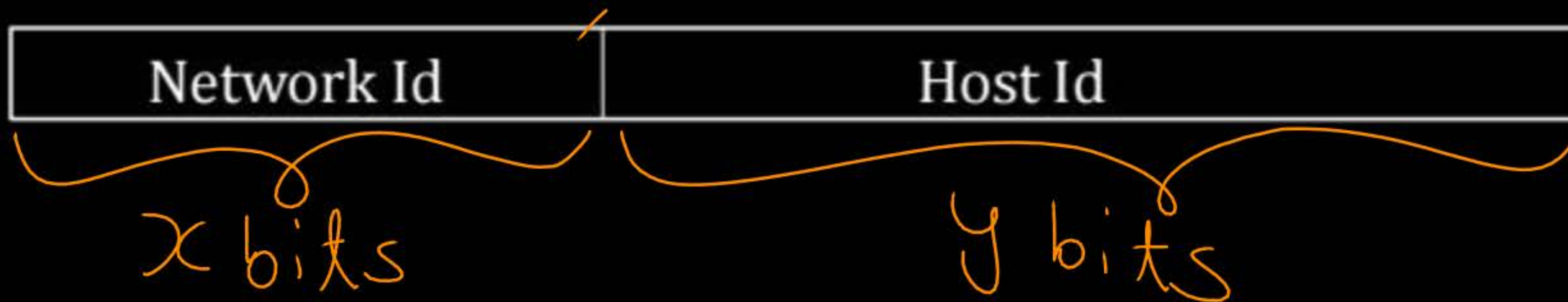
Topic : IP Address



→ IP address having two sections :

1. Network Identifier (Net ID) : x bits
2. Host Identifier (Host ID) : y bits

→ Size of IP address field = (x + y) bits





Topic : Network Identifier

- Unique network identifier (x bits)
[Range : 0 to $(2^x - 1)$]
- Assigned to each connected network
- Managed by IANA
[Internet Assigned Numbers Authority]
- Total number of networks (worldwide) can be : 2^x



Topic : Host Identifier



→ Unique host identifier (y bits) per network
[Range : 0 to $(2^y - 1)$]

→ Assigned to each connected host in the network



Topic : IP Protocol



Two IP versions :

1. IPv4

→ IPv4 Address (32 bits)

→ 4 bytes [$x + y = 32$]

2. IPv6

→ IPv6 Address (128 bits)

→ 16 bytes [$x + y = 128$]



Topic : IPv4 Address Representation

→ Size of IPv4 Address field = 32 bits

→ Two representation of IPv4 address :

1. Binary Representation [32 bits]

101101 0101

2. Dotted Decimal Notation

(32 bit)

Human Readable



Topic : Dotted Decimal Notation

→ 4 Octet, separated by period (.)

→ P . Q . R . S

Ex:- 202.12.135.79

→ Range value of every octet is 0 to 255

[8-bit Range : 0 to $(2^8 - 1)$]

0 to 255

Total 2^8 numbers

$$0 \leq P, Q, R, S \leq 255$$



Topic : Binary Code

Base-2



→ Weighted Code

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
b_7	b_6	b_5	b_4	b_3	b_2	b_1	b_0
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>



Topic : Binary Code



$$00000000 = \boxed{0}$$

$$00000001 = \boxed{1}$$

$$00000010 = \boxed{2}$$

$$00000100 = 4$$

$$00001000 = 8$$

$$00010000 = 16$$

$$00100000 = 32$$

$$01000000 = 64$$

$$10000000 = 128$$

$$10000000 = \boxed{128}$$

$$\underline{1}\underline{1}000000 = 192 = 128 + 64$$

$$\underline{1}\underline{1}\underline{1}00000 = 224 = 192 + 32$$

$$\underline{1}\underline{1}\underline{1}\underline{1}0000 = 240 = 224 + 16$$

$$\underline{1}\underline{1}\underline{1}\underline{1}\underline{1}000 = 248 = 240 + 8$$

$$\underline{1}\underline{1}\underline{1}\underline{1}\underline{1}\underline{1}00 = 252 = 248 + 4$$

$$\underline{1}\underline{1}\underline{1}\underline{1}\underline{1}\underline{1}\underline{1}0 = 254 = 252 + 2$$

$$11111111 = 255 = (2^8 - 1)$$



Topic : Binary Code



$$\begin{aligned}00000001 &= \boxed{1} \\000000\underline{11} &= 3 = 2 + 1 \\00000\underline{111} &= 7 = 4 + 3 \\0000\underline{1111} &= 15 = 8 + 7 \\000\underline{11111} &= 31 = 16 + 15 \\00\underline{111111} &= 63 = 32 + 31 \\0\underline{1111111} &= 127 = 64 + 63 \\1\underline{1111111} &= 255 = 128 + 127\end{aligned}$$

Example 1 :-

[MCQ]



#Q. Consider given IPv4 address in binary representation, identify correct decimal representation of it?

01010101 11011011 10100101 01110111
85 219 165 119

☒ A 85.219.164.119

☒ B 85.218.165.119

☒ C 85.219.165.118

☒ D 85.219.165.119

Ans: D

Example 2 :-

[MCQ]



#Q. Consider given IPv4 address in decimal representation, identify correct binary representation of it?

200.100.150.50

$$200 = 11001000$$

$$100 = 01100100$$

$$50 = 00110010$$

$$150 = 10010110$$

$$150 = 10010110$$

☒ A 11001000 01100100 10010110 00110110

☐ B 11001000 01100100 10010100 00110010

☒ C 11001000 01100100 10010110 00110010

☒ D 11001000 01100110 10010110 00110010

Ans: C



Topic : IPv4 Address



Problem : How many bits are assigned for network id field ?

$NetID = x \text{ bits}$

→ IPv4 address are categorized into two type :

1. Classfull IPv4 Address

→ Static Assignment : NetID field size defined implicitly $[x = 8 \text{ or } 16 \text{ or } 24]$

2. Classless IPv4 Address ✓

→ Dynamic Assignment : Need to define NetID field size explicitly

$x \rightarrow \text{vary}$



Topic : Classless IPv4 Address

- Number of bits for network id field are assigned 'Dynamically'
- Classless IPv4 Address : P.Q.R.S / x
↑
- Prefix / Slash notation [x] represent "number of network id bits"
↑

$$\text{NetID} = x \text{ bits}$$

$$\text{HostID} = y \text{ bits} = (32 - x) \text{ bits}$$



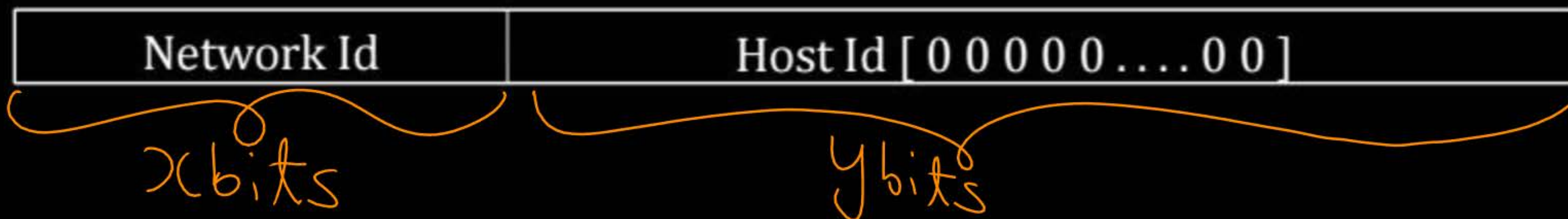
Topic : Network Address

V. Imp.

- Special IP address
- It can not be an IP address of any host in the network
- Used to represent a network

NetID field = As Assigned

HostID field = All Zero Bits



Example 3 :-

[MCQ]



#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "1100 1100 1010", identify correct decimal representation of network address (special IPv4 address) which is used to represent this network?

☐ A 204.144.0.0

☒ B 204.160.0.0

☐ C 212.144.0.0

☐ D 212.160.0.0

Ans: B

Network Address =

11001100 10100000 00000000 00000000

12 bit NetID 20 bit HostID

204.160.0.0



Topic : Modes of Transmission

→ Three modes of IP transmission : (in IPv4)

1. Unicast
2. Broadcast
3. Multicast



2 mins Summary



Topic

Network Address ✓

Topic

~~Broadcast Address~~



THANK - YOU

